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Train a custom object detection model with MediaPipe Model Maker

In this colab notebook, you'll learn how to use [MediaPipe Model Maker]() to train a custom object detection model to detect dogs.

The Model Maker library uses *transfer learning* to simplify the process of training a TensorFlow Lite model using a custom dataset to use with the [MediaPipe Object Detector task]().

Preparation

Install the required packages

Start by installing the required packages, including the Model Maker package from the [GitHub repo]() and the pycocotools library you'll use for evaluation.

Start coding or generate with AI.

```
!pip install --upgrade pip
!pip install 'keras<3.0.0' mediapipe-model-maker
```

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Resources

You are subscribed to Colab Pro.
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Available: 143.81 compute units

Usage rate: approximately 0 per hour

You have 0 active sessions.

Manage session

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Downloading portalocker-3.0.0-py3-none-any.whl (19 kB)
Building wheels for collected packages: seqeval
  Building wheel for seqeval (setup.py) ... done
  Created wheel for seqeval: filename=seqeval-1.2.2-py3-none-
  Stored in directory: /root/.cache/pip/wheels/1a/67/4a/ad408
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Installing collected packages: wrapt, typeguard, tensorflow-m
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    Found existing installation: wrapt 1.17.0
    Uninstalling wrapt-1.17.0:
      Successfully uninstalled wrapt-1.17.0
  Attempting uninstall: typeguard
    Found existing installation: typeguard 4.4.1
    Uninstalling typeguard-4.4.1:
      Successfully uninstalled typeguard-4.4.1
    Attempting uninstall: ml-dtypes
```

Import the required packages.

```
import os
import tensorflow as tf
assert tf.__version__.startswith('2')
from google.colab import files

from mediapipe_model_maker import object_detector
```

```
/usr/local/lib/python3.10/dist-packages/tensorflow_addons/uti

TensorFlow Addons (TFA) has ended development and introductio
TFA has entered a minimal maintenance and release mode until 
Please modify downstream libraries to take dependencies from 

For more information see: https://github.com/tensorflow/addon

  warnings.warn(
```

Prepare the dataset

This dataset contains about 130 images of 3 different dogs: a Golden Retriever, a Labrador Retriever mix, and a brindle-colored Formosan Mountain Dog. This is an example image from the dataset.

We start with downloading the dataset.

```
#!wget https://storage.googleapis.com/mediapipe-assets/dogs2.zip
!unzip rocket8.zip
train_dataset_path = "rocket8/train"
validation_dataset_path = "rocket8/validate"
```

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      inflating: rocket8/train/Annotations/img301.xml
```

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  inflating: rocket8/train/Annotations/img114.xml  
```

Train the object detection model

Step 1: Load the dataset

- Images in `train_data` is used to train the custom object detection model.
- Images in `val_data` is used to check if the model can generalize well to new images that it hasn't seen before.

```
train_data = object_detector.Dataset.from_pascal_voc_folder(
    'rocket8/train',
    cache_dir="/tmp/od_data/train",
)

val_data = object_detector.Dataset.from_pascal_voc_folder(
    'rocket8/validate',
    cache_dir="/tmp/od_data/validatation")
```

Step 2: Train the TensorFlow model with the training data.

- Set `batch_size = 8` here so you will see that it takes 13 steps to go through the 139 images in the training dataset.
- Set `learning_rate = 0.3` here so the model will adjust its weights at a rate of 0.3.
- Set `epochs = 50`, which means it will go through the training dataset 50 times. You can look at the validation accuracy during training and stop when you see validation loss (`val_loss`) stop decreasing to avoid overfitting.

```
hparams = object_detector.HParams(batch_size=8, learning_rate=0.3,
options = object_detector.ObjectDetectorOptions(
    supported_model=object_detector.SupportedModels.MOBILENET_V2,
    hparams=hparams
)
model = object_detector.ObjectDetector.create(
    train_data=train_data,
    validation_data=val_data,
    options=options)
```

```
/usr/local/lib/python3.10/dist-packages/keras/src/engine/func
  inputs = self._flatten_to_reference_inputs(inputs)
WARNING:tensorflow:`tf.keras.layers.experimental.SyncBatchNor
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Model: "retina_net_model"
_____________________________________________________________
 Layer (type)                Output Shape              Param
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 mobile_net (MobileNet)      {'2': (None, 64, 64, 24   225798
                             ),
                              '3': (None, 32, 32, 32
                             ),
                              '4': (None, 16, 16, 96
                             ),
                              '5': (None, 8, 8, 320)
                             , '6': (None, 8, 8, 128
                             0)}

 fpn (FPN)                   {'5': (None, 8, 8, 128)   149056
                             , '4': (None, 16, 16, 1
                             28),
                              '3': (None, 32, 32, 12
                             8),
                              '6': (None, 4, 4, 128)
                             , '7': (None, 2, 2, 128
                             )}

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 retina_net_head (RetinaNet  ({'3': (None, 32, 32, 8   179189
 Head)                       1),
                              '4': (None, 16, 16, 81
                             ),
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                              '6': (None, 4, 4, 81),
                              '7': (None, 2, 2, 81)}
                             , {'3': (None, 32, 32,
                             36),
                              '4': (None, 16, 16, 36
                             ),
                              '5': (None, 8, 8, 36),
                              '6': (None, 4, 4, 36),
                              '7': (None, 2, 2, 36)}
                             , {})

=============================================================
Total params: 2586229 (9.87 MB)
```

```
Trainable params: 2540597 (9.69 MB)
Non-trainable params: 45632 (178.25 KB)
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Epoch 1/50
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```

Step 3. Evaluate the model with the validation data.

After training the object detection model using the images in the training dataset, use the 27 images in the validation dataset to evaluate how the model performs against new data it has never seen before.

The evaluation metrics are same as COCO.

```
loss, coco_metrics = model.evaluate(val_data, batch_size=4)
print(f"Validation loss: {loss}")
print(f"Validation coco metrics: {coco_metrics}")
```

```
17/17 [==============================] - 3s 91ms/step - total_
creating index...
index created!
creating index...
index created!
Running per image evaluation...
Evaluate annotation type *bbox*
DONE (t=0.31s).
Accumulating evaluation results...
DONE (t=0.12s).
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 Average Precision  (AP) @[ IoU=0.50      | area=   all | max
 Average Precision  (AP) @[ IoU=0.75      | area=   all | max
 Average Precision  (AP) @[ IoU=0.50:0.95 | area= small | max
 Average Precision  (AP) @[ IoU=0.50:0.95 | area=medium | max
 Average Precision  (AP) @[ IoU=0.50:0.95 | area= large | max
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 Average Recall     (AR) @[ IoU=0.50:0.95 | area=medium | max
 Average Recall     (AR) @[ IoU=0.50:0.95 | area= large | max
Validation loss: [0.7392598986625671, 0.4557432532310486, 0.0
Validation coco metrics: {'AP': 0.39894685, 'AP50': 0.8752042
```

Step 4: Export as a TensorFlow Lite model.

After creating the model, convert and export it to a Tensorflow Lite model format, and then download it for later use in an on-device application with MediaPipe Tasks.

```
model.export_model('rocket8.tflite')
!ls exported_model
files.download('exported_model/rocket8.tflite')
```

```
Exporting a floating point model
/usr/local/lib/python3.10/dist-packages/keras/src/engine/funct
  inputs = self._flatten_to_reference_inputs(inputs)
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